ASSIGNMENT No. 15

Problem Statement: Write a C program to generate pseudo random numbers.

Objectives:
- To understand the concept of pseudo random nos
- To learn the use of rand() & srand() functions in C
- To generate random numbers using the standard library.
- To understand the role of header files in C programming.

Theory: In C programming pseudo random numbers are generated using the rand() function available in the stdlib.h header file. The numbers are not truly random but are generated using a deterministic algorithm. The srand() function is used to set the seed value for random no. generation. Usually, the current time is used as a seed to produce different sequences of nos each time the program ~~time~~ runs.

Platform: Linux

Input: No. of random numbers to be generated

Output: Displays generated pseudo random nos.

Sample Input: Enter how many random no. you want: 5

DOMS Page No.
Date / /

Sample Output:
Pseudo Random Numbers:
183746
928374
567382
345678
987234

Conclusion: Thus, the C program successfully generates pseudo random numbers using the rand() function and srand() for seeding.

FAQs:

1. What are pseudo random ~~rando~~ numbers in C?
A. Pseudo-random numbers are nos. that look random but are actually generated using a mathematical formula. They follow a fixed pattern. If you use the same starting value, you get the same sequence everytime.

2) Which function is used to generate random numbers in C?
A. The function used is: rand()
It generates a random integer between 0 & RAND_MAX (a constant value)

3) Which header file is required for rand() function?
A. You need to include:
#include <stdlib.h>

4) What is the purpose of srand() function?

A. srand(seed):
It is used to initialized the random number generator.
The seed determines the starting pt of the random sequence.
If you don't use srand(), the program will generate the same numbers every time.

5) Why is time() used with srand()?

A. srand(time(0)):
1) time() gives the current time.
2) Since time changes every second, it provides a different seed each time the program runs.
3) This makes the random numbers different in every execution.